Exercises on Arrays

Computer Programming 1

2026-10-13

Outline

Data I/O
Write two functions:

- the first reads a mono-dimensional array of numbers from the user
- the second outputs the array

Assume the dimension is set a priori, tell the user how many number have to be entered.

Data I/O, multi-dimensional arrays

Write two functions:

- the first reads a matrix (two D array) of numbers from the user
- the second outputs the matrix

Assume the dimensions are set a priori, enter the data row by row (e.g., "Enter 5 numbers for row 1")

Hints and Challenges

- What happens if the user enters 7 values in a row?
- Can you fix the input so that these kind of problems don't happen?
- Use setw (from iomanip) to display the matrix in a nicely-formatted way
- Write the functions so that you can reuse them in multiple programs (in principle, they could become the part of a library)

Sum

- Write a function which computes the sum of all the elements on an array (Hint: use a support variable which keeps the sum)

Maximum

- Write a function which computes the sum of all the elements on an array (Hint: use a support variable which keeps the maximum found so far. How do you initialize it in a ways wihch is good for any array?)

Moving average

The moving average of a series is computed by averaging a pre-defined number of contiguous elements.

For instance, if I decide to compute the moving average using three elements:

- the moving average can be be computed starting from the third element (since, before that, I don't have enough elements for computing it)
- the moving average of the third element is the average of the first, second, and third element
- the moving average of the fourth element is the average of the fourth, third, and second element
- ... and so on.

Swap

- Write a function which, for each index of an array swaps the element at index i with the element at index i $+$ 1 if the value at position i$+1$ is lower than the value at position i
- Extend the previous exercise so that the process is repeated until there are no more swaps.
- Print the output array. What happens?

Sieve of Eratosthenes

In mathematics, the sieve of Eratosthenes is an ancient algorithm for finding all prime numbers up to any given limit.

- Suppose you want to find the prime numbers between 1 and N
- You create an array of N elements, which contains the numbers from `1` to `n` (e.g., `a[n] = n`).
- The idea is that you "delete" (set to `-1`) all numbers which are not prime

How

- You initialize an index `i` to 2 (a prime number). You visit the array `i` by `i` and set all the elements to `-1`. This "deletes" from the array all elements which are multiple of 2 (not prime)
- Now you move `i` to the next element different from `-1` and continue

Example

Start:

1 2 3 4 5 6 7 8 9 10

We set `i = 2` and set to −1 the elements at multiple of two:

1 2 3 -1 5 -1 7 -1 9 -1

We now move `i` to the next element in the array which is not −1 (that is, 3) and set to −1 all elements at positions multiple of 3 (6, 9):

1 2 3 -1 5 -1 7 -1 -1 -1

we now move `i` to 5 (the first number which is not −1) and continue

Tip

Use two nested loops:

- the first moves `i` to the first element of the array which is different from `-1`
- the second deletes the elements from the array

Drawing board

- An image can be represented by a 2D array of integers
- A common dimension is 1920x1024
- In a BW drawing, we can represent white with 255 and black with 0

Example: Rectangle

255	255	255	255	255	255	255	255	255	255
255	0	0	0	0	0	0	0	0	255
255	0	255	255	255	255	255	255	0	255
255	0	0	0	0	0	0	0	0	255
255	255	255	255	255	255	255	255	255	255

Exercise

- Initialize a drawing board 1920x1024 with white
- Ask the user the coordinates of a rectangle to draw, in the form top, left corner, width and height
- Draw the rectangle (by setting the appropriate cells to black) and output it

5x5

- 5x5 is a game where you need to put 5 cells in a row
- It is played on a big board (for instance, 20x20) by two players
- A cell can be empty (0), occupied by black (1), or occupied by white (2)
- Write a program which checks whether a given board (with some cells occupied) contains a winning sequence for black or white
  - test first horizontally, then vertically
  - challenge: test for diagonals

Hint

- Create the main loops (each made of two cycles):
  - first loop will check horizontal lines (outer loop for one row, inner loop for rows)
  - first loop will check vertical lines (outer loop for one column, inner loop for columns)
- Keep two variables, the first keeping the count of consecutive white cells founds and the second keeping the coung of consecutive black cells found. They start at zero

Challenge

- Option 1: you "sweep" all diagonals
- Option 2: you find the cell with a 1. Every time you find one, you start moving diagonally, to check whether there is match (kind of inefficient)